

8. Vanadium is a transition element that forms compounds with a wide range of oxidation states. Its compounds are also used in a number of catalysts.

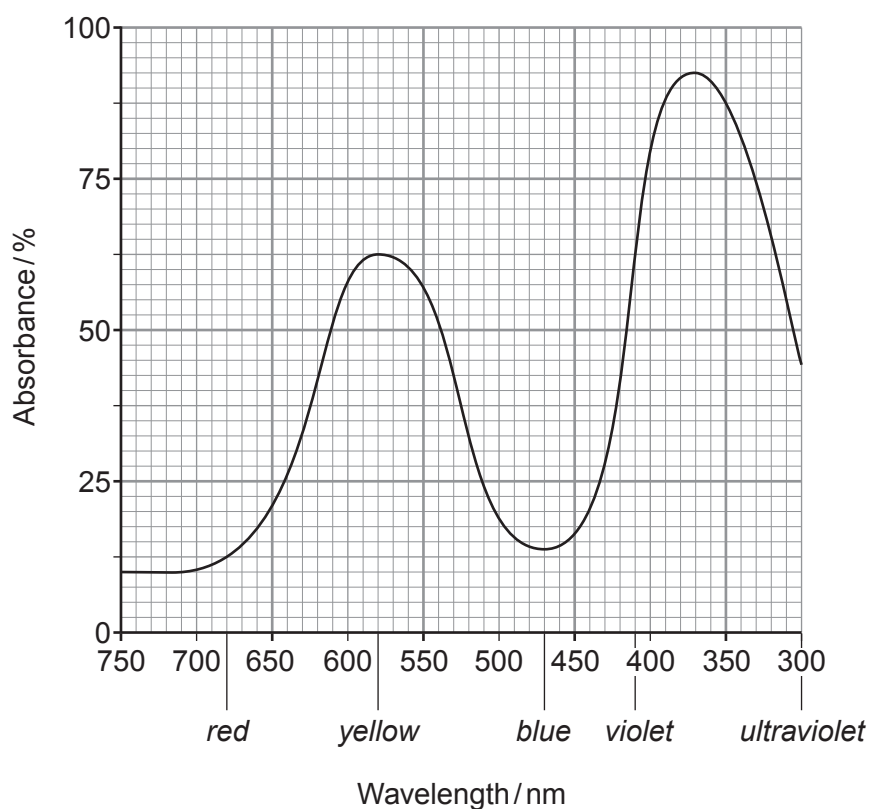
(a) State why transition elements can form a range of oxidation states in their compounds.

[1]

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- (b) The visible spectrum of an aqueous solution of a compound containing V^{3+} ions is shown below.



- (i) Suggest the colour expected for aqueous solutions of V^{3+} compounds.

Give a reason for your answer.

[2]

- (ii) In aqueous solutions, most vanadium ions form complexes with water acting as a ligand.

I. Give the meaning of the term ligand.

[1]

II. Describe the bonding that occurs between the ligand and the transition metal ion.

[2]

- (c) VO_2^+ ions can be reduced to VO^{2+} ions by iodide ions as shown below.



- (i) Write the half-equation for the oxidation of iodide ions.

[1]

- (ii) Write the half-equation for the reduction of VO_2^+ ions in acid solution.

[1]



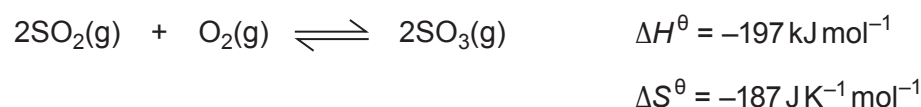
(d) Vanadium(V) oxide, V_2O_5 , is used as a heterogeneous catalyst in the contact process.

(i) State what is meant by the term heterogeneous in this context. [1]

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(ii) This catalyst is used to catalyse the oxidation of sulfur dioxide.



I. The standard enthalpy change of formation of sulfur dioxide is -297 kJ mol^{-1} . Calculate the standard enthalpy change of formation of sulfur trioxide, SO_3 . [2]

$$\Delta_f H^\theta = \dots\dots\dots \text{ kJ mol}^{-1}$$

II. A student calculates the value at which $\Delta G = 0$ and states that this is the minimum temperature needed for the reaction to occur.

Find the temperature at which $\Delta G = 0$ and state, giving a reason, whether this is the **minimum** temperature needed for the reaction to occur. [3]

$$\text{Temperature} = \dots\dots\dots \text{ K}$$

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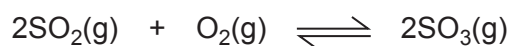
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- III. Samples of 0.040 mol of SO_2 and 0.040 mol of O_2 are placed in a sealed vessel of volume 2.00 dm^3 . The reaction is allowed to come to equilibrium, giving a concentration of $0.014 \text{ mol dm}^{-3}$ of SO_3 .

Calculate the value of the equilibrium constant, K_c , under these conditions, giving its unit. [4]



$K_c =$

Unit

